

RISHABH SABHARWAL

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EDUCATION

MSc in Artificial Intelligence, University of Edinburgh, UK (2026) | **Grade: Distinction***

B.Tech in ECE (AI & ML), NSUT, India (2024) | **GPA: 3.68/4** (CVSPK Fellow, **Top 5%**)

SELECTED PUBLICATIONS

A.Mazumder, **Rishabh Sabharwal**, ..., P. Rathore, 'On Convergence of Adam with Data Dependent Stepsize' in **IEEE Transactions on AI** (Q1 Journal, Impact Factor: 4.8). ([Link](#))

K. Annuranjana, N. Shrivastava*, A. Johar*, **Rishabh Sabharwal***, ..., 'Does Reasoning Kill the Joke? Long-Context Humor Understanding in Hindi' in **C3NLP Workshop, ACL 2026**.

Rishabh Sabharwal, H. Wang, A. Storkey, J. Pan, 'Multi-turn Evaluation of Deep Research Agents' (Pre-print).

SELECTED PROJECTS

Multi-turn Evaluation of Deep Research Agents (MSc Thesis - In Progress)

- Designed a multi-turn evaluation framework to assess agents' ability to iteratively refine reports; benchmarked GPT-4.1-mini, GPT-4.1 via OpenAI API with prompt optimization for evaluation and feedback generation.
- Developed a multimodal probe to benchmark agents' ability to extract and embed relevant images in reports.

LLM Post-Training on Small-Scale Models: ([Code](#))

- Fine-tuned Qwen2.5-0.5B on GSM8K via SFT and RLHF; mitigated format reward over-optimisation by using a **Process Reward Model** assigning dense rewards to CoT steps, achieving **37% accuracy** on test set.
- Designed a **latent-space knowledge distillation** framework transferring K latent reasoning tokens from Qwen2.5-7B directly into the student's representation space, achieving **41% accuracy** on the test set.

FastAPI Safety Gateway: ([Code](#))

- Built a modular guardrail pipeline around a vLLM-served Llama-3.1-8B-Instruct exposed as a REST endpoint, incorporating toxicity/harm classification, prompt injection detection, and PII detection/anonymization.
- Developed an **agentic safety loop** that rewrites risky prompts and repairs unsafe outputs via policy-conditioned re-prompting, with user-adjustable thresholds and transparent decision tracing.

EXPERIENCE

ML Engineer

Indian Institute of Science

Jan 2024 - Jul 2025

Bengaluru, India

- Built **large-scale graph** algorithms for fraud detection on billion-node graphs in partnership with NPCI.
- Developed ReGA, a parameter-free spectral GNN module for global context filtering; achieved SOTA on 20 real-world datasets, reducing parameters by **55x** and GFLOPs by **100x**.
- Proposed a network and data-dependent constant learning rate to address decaying scheduler limitations; demonstrated faster convergence on both stochastic and deterministic Adam variants. ([Paper](#))

RELEVANT COURSEWORK

- Advanced NLP** (Transformers, PEFT, Knowledge Distillation, RAGs, Information Retrieval, Multimodal LLMs, LLM Agents), **ML Systems** (GPU Programming, LLM Inference Optimisation), **Probabilistic Modelling & Reasoning** (Bayesian Inference, Statistical ML, Optimisation Theory)

SKILLS

- Programming & Frameworks:** Python, Pandas, NumPy, PyTorch, Triton, FastAPI
- ML/LLM Tooling:** Hugging Face, PEFT (LoRA, QLoRA), vLLM, Weights & Biases, GCP (Vertex AI)
- Agentic AI:** LangChain, LangGraph, Multi-Agent Systems
- DevOps/Systems:** Git, Linux, CI/CD, Distributed Training (PyTorch DDP)